

ME 605 | Computational Fluid Dynamics

Project 2

Due: 11:59 pm, 19 October 2025

Instructions

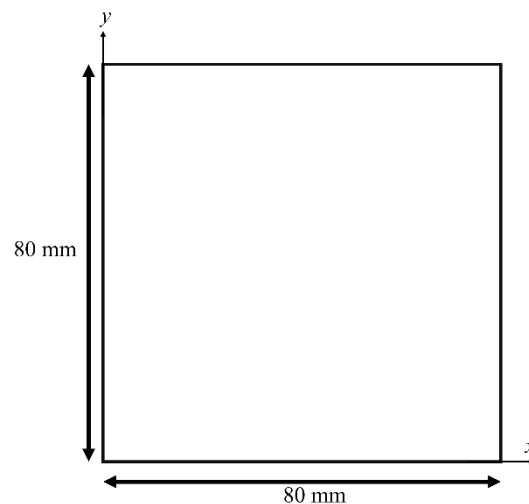
1. You can choose any programming language.
2. Do not use any in-built or intrinsic functions. You are expected to write your own computer program (including solving the system of algebraic equations) and a project report
3. Discussion among students is permitted.
4. Your report must consist of (1) problem statement, (2) mesh details and approach for discretization; (3) derivation and presentation of the final form of the discretized equations; (4) solution methodology, (5) results and discussion; (6) concluding remarks. Note that an in-depth analysis and discussion of results is required.
5. Report must be prepared using WORD or LaTeX. Handwritten reports will not be accepted.
6. Submit the project report and the code using the link/form that will be provided before the submission deadline.
7. Usage of any artificial intelligence (AI) tools to write project reports or develop code modules is a violation of the course policy

Project Statement: Computer Solution of Parabolic PDEs

You are required to write a computer program and solve the following 2D unsteady heat conduction equation using the finite difference method:

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

on a domain shown below:



The initial temperature of the domain is 300 K. The boundary conditions are as follows:

$$T(x, y = 0) = 323 \text{ K}$$

$$T(x, y = 80 \text{ mm}) = 423 \text{ K}$$

$$T(x = 80 \text{ mm}, y) = 473 \text{ K}$$

$$h(T_{\infty} - T(x = 0, y)) = -\lambda \left. \frac{\partial T}{\partial x} \right|_{x=0}$$

$$h = 250 \frac{\text{W}}{\text{m}^2\text{K}}, \quad \lambda = 5 \frac{\text{W}}{\text{m K}}, \quad T_{\infty} = 573 \text{ K} \text{ and } \alpha = 5 \times 10^{-6} \frac{\text{m}^2}{\text{s}}$$

Where h , λ , T_{∞} and α are convective heat transfer coefficient, thermal conductivity, free stream temperature, and thermal diffusivity, respectively.

Note that a first-order numerical approximation to the Neumann boundary condition may be necessary to preserve the tridiagonal structure of the coefficient matrix.

- (1) Compute the time evolution of the temperature field and the steady-state temperature field using the explicit scheme for an 81×81 grid. What is the critical time step (Δt_{crit}) beyond which the method becomes unstable? Plot the temperature variation along the centerlines ($x = 40 \text{ mm}, y$) and ($x, y = 40 \text{ mm}$) separately at six time instances spaced equally apart between $t = 0$ to $t = t_{end}$. Here, t_{end} refers to the time instant corresponding to the attainment of the steady-state condition. Please plot temperature variations for all six time instances in the same plot.
- (2) Compute the time evolution of the temperature field and the steady-state temperature field using the implicit scheme for an 81×81 grid. Take $\Delta t_{CFD} = 2\Delta t_{crit}$. Plot the temperature variation along the centerlines ($x = 40 \text{ mm}, y$) and ($x, y = 40 \text{ mm}$) separately at six time instances spaced equally apart between $t = 0$ to $t = t_{end}$. Here, t_{end} refers to the time instant corresponding to the attainment of the steady-state condition. Please plot temperature variations for all six time instances in the same plot.
- (3) Compute the time evolution of the temperature field and steady-state temperature field using the Crank-Nicolson scheme for an 81×81 grid. Plot the temperature variation along the centerlines ($x = 40 \text{ mm}, y$) and ($x, y = 40 \text{ mm}$) separately at six time instances spaced equally apart between $t = 0$ to $t = t_{end}$. Here, t_{end} refers to the time instant corresponding to the attainment of the steady-state condition. Please plot temperature variations for all six time instances in the same plot.

You can use any iterative method used in the previous project to solve the system of linear equations.